

USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech (EVT) (CBCS) Degree Examinations

December 2024 – January 2025

22EVT111 – ADVANCED BATTERY TECHNOLOGY FOR ELECTRIC VEHICLES

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Illustrate the major components of EV and the battery cell basics.	6	L*2	1	2
b)	Summarize the four steps of conventional processing method for manufacturing the AB5 battery materials.	4	L2	1	2
c)	Define the state of charge (SOC). Explain in detail the practical means of calculating the SOC.	10	L2	2	2
2. a)	What are the characteristics of the Li-ion battery that are favorable for EV applications.	6	L2	1	2
b)	Discuss the functioning of solid-state Lithium Polymer batteries.	4	L3	1	2
c)	Explain the 4 types of battery capacity tests.	10	L2	2	2
3. a)	Briefly explain the chemical functioning of fuel cell.	8	L2	1	2
b)	What are the factors influencing the energy balance of the EV, Explain?	12	L2	2	2
Unit – II					
4. a)	List and briefly explain the Fast-charging prerequisites.	5	L2	3	2
b)	Explain the three methods of charge termination during the battery charging process, to monitor the cell temperature.	6	L2	3	2
c)	Enumerate the observations made by the cold weather performance on a compact size EV using VRLA batteries.	9	L2	4	2
5. a)	Explain how the VRLA batteries are best charged using a multistep charging algorithm.	8	L2	3	2
b)	Discuss the limitations of fast charging.	4	L3	3	2
c)	Rephrase the set of initial conditions that must be satisfied to successfully perform range tests on the EV.	8	L2	4	2
6. a)	Explain the two primary coupling methods of transferring power to EVs.	8	L2	3	2
b)	Enumerate the steps involved in EV speedometer recalibration.	6	L2	4	2
c)	Explain how the Inductive charging makes the recharging easier by transferring electrical energy from the electric utility charge port to the EV battery pack.	6	L3	4	2
Unit – III					
7. a)	Explain the main components of a BPMS.	4	L2	5	2
b)	Describe the model of BPMS.	8	L2	5	2
c)	Illustrate the BPMS charge protector and charge indicator.	8	L2	5	2
8. a)	Illustrate the design analysis of battery thermal management system.	10	L2	5	2
b)	Explain the smart battery and BPMS diagnostics control.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
